Christoph Gerhardt

Dr.-Ing. Candidate — Computer Vision & Deep Learning

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Google Scholar *Web of Science*

Research Interests

Computer Vision, Semantic Segmentation, 360° Image Processing, Weather-Robust Perception, Data Augmentation, Outdoor Scene Understanding, Neural Network Architectures, Object Detection, Panoramic Imaging

Education

2019 – 2026 (exp.) **Dr.-Ing. (PhD) in Computer Science**, *Technische Universität Ilmenau*, Ilmenau, Germany
Thesis: *Comprehensive Outdoor Scene Understanding Through Multi-Scale Neural Networks: From 360° Object Detection to Panoramic Segmentation and Weather-Robust Augmentation*
Supervisor: Prof. Dr. rer. nat. Wolfgang Broll
Research Group: Virtual Worlds and Digital Games (VWDG)

2014 – 2017 **M.Sc. Computer Engineering (*Ingenieurinformatik*)**, *Technische Universität Ilmenau*, Ilmenau, Germany, *Grade: 1.5 (very good)*
Specialisation: Multimedia Information and Communication Systems
Thesis: *Selektive Verschlüsselung von Gesichtern in Videodaten unter Verwendung des H.264-Standards*

2010 – 2014 **B.Sc. Computer Engineering (*Ingenieurinformatik*)**, *Technische Universität Ilmenau*, Ilmenau, Germany, *Grade: 1.8 (good)*
Thesis: *Entwicklung eines Verfahrens zur Verbesserung von Mikrofonklassifizierung basierend auf der Analyse von Umwelteinflüssen*

2009 **Abitur**, *Gymnasium "Am Lindenberg"*, Ilmenau, Germany, *Grade: 1.5*

Professional Experience

Feb 2019 – present **Research Associate (*Wissenschaftlicher Mitarbeiter*)**, *Technische Universität Ilmenau*, Ilmenau, Germany
Virtual Worlds and Digital Games Group (Prof. Broll)
- Research on neural-network-based semantic segmentation of outdoor scenes
- Development of weather-robust augmentation frameworks (AWARE / SWIFT / PRISM / PROVE)
- 360° panoramic image analysis for traffic sign detection and sky/cloud segmentation
- Co-supervision of student theses; teaching duties in VR/AR and computer vision
- Collaboration with NOVASIB GmbH on road inventory management (TT-SIB INFOSYS®)

Feb 2017 – Jan 2019 **Software Developer**, *Fraunhofer Institute for Digital Media Technology (IDMT)*, Ilmenau, Germany
Group: Virtual Acoustics
- Software development for spatial audio processing and rendering systems

Jul 2016 – Jan 2017 **Student Research Assistant (*Hilfswiss. Mitarbeiter*)**, *Fraunhofer IDMT*, Ilmenau, Germany
Group: Media Distribution and Security

Internships

Sep 2013 – Jan 2014 **Research Intern**, *Fraunhofer IDMT*, Ilmenau, Germany
Group: Media Distribution and Security

Mar 2013 **Intern**, *SIOS Meßtechnik GmbH*, Ilmenau, Germany

Sep – Oct 2012 **Intern**, *GPM Anlagenbau und Industrieanlagen GmbH*, Mainz, Germany

Civic Service

2009 – 2010 **Voluntary Military Service (*Freiwilliger Wehrdienst*)**, *Führungsunterstützungsbataillon 292*, Dillingen a. d. Donau, Germany

Publications

Peer-Reviewed — First Author (Dissertation-Related)

[1] **C. Gerhardt** and W. Broll, "SkyCloud360: Sky and Cloud Segmentation in Equirectangular Images," in *Proc. 10th ICIVC*, Chengdu, China, 2025, pp. 48–58. doi:10.1109/ICIVC66358.2025.11200399.

[2] **C. Gerhardt**, F. Weidner, and W. Broll, "SkyCloud: Neural Network-Based Sky and Cloud Segmentation from Natural Images," in *Proc. 8th ICIVC*, Dalian, China, 2023. doi:10.1109/ICIVC58118.2023.10270450.

[3] **C. Gerhardt**, F. Weidner, and W. Broll, "OUTSIDE: Multi-Scale Semantic Segmentation of Universal Outdoor Scenes," in *Proc. IEEE 23rd MMSP*, Tampere, Finland, 2021, pp. 1–6. doi:10.1109/MMSP53017.2021.9733652.

[4] **C. Gerhardt** and W. Broll, "Neural Network-Based Traffic Sign Recognition in 360° Images for Semi-Automatic Road Maintenance Inventory," in *Proc. IEEE 23rd ITSC*, 2020, art. no. 9294610. doi:10.1109/ITSC45102.

[5] **C. Gerhardt**, M. M. Salman, and W. Broll, "AWARE: Augmentation for Weather-Adverse Robustness Evaluation," *IEEE Access*, 2025 *(under review)*.

Peer-Reviewed — Co-Author

[6] S. Arévalo Arboleda *et al.* incl. **C. Gerhardt**, "Robot, Avatar, or Human: The Impact of Partner Representation and Task on the Communication Experience," *Proc. ACM CSCW*, vol. 9, art. CSCW472, 2025. doi:10.1145/3757653.

[7] E. Makled, **C. Gerhardt**, T. Schwandt, F. Weidner, and W. Broll, "Evaluating Behavioral Realism in AR and VR. . . ," *The Visual Computer*, vol. 41, no. 14, pp. 11721–11733, 2025. doi:10.1007/s00371-025-03934-5.

[8] V. Mikhailova *et al.* incl. **C. Gerhardt**, "Addressing the Avatar in the Room. . . ," in *ITAP/HCII 2025*, LNCS 15809, Springer, 2025. doi:10.1007/978-3-031-92707-2_21.

[9] V. Mikhailova *et al.* incl. **C. Gerhardt**, "Work-in-Progress: Older Adults' Experiences with an AR Communication System," in *Proc. ACM IMX*, Stockholm, Sweden, 2024. doi:10.1145/3639701.3663641.

[10] E. Makled, **C. Gerhardt**, T. Schwandt, F. Weidner, and W. Broll, "Investigating Behavioral Realism of Single-Point IK Animated Avatars. . . ," in *Proc. Int. Conf. Cyberworlds*, Kofu, Japan, 2024, pp. 1–8. doi:10.1109/CW64301.2024.00010.

[11] F. Weidner *et al.* incl. **C. Gerhardt**, "Eyes on the Narrative: Exploring the Impact of Visual Realism and Audio Presentation on Gaze Behavior in AR Storytelling," in *Proc. ACM ETRA*, Glasgow, UK, 2024. doi:10.1145/3649902.3653344.

[12] V. Mikhailova, **C. Gerhardt** *et al.*, "Age and Realism of Avatars in Simulated AR," in *Proc. IEEE VR*, Orlando, FL, 2024, pp. 83–93. doi:10.1109/VR58804.2024.00032.

[13] S. Arévalo Arboleda *et al.* incl. **C. Gerhardt**, "Beyond Looks: Agent Movement and Audiovisual Spatial Coherence in AR," in *Proc. IEEE VR*, Orlando, FL, 2024, pp. 502–512. doi:10.1109/VR58804.2024.00071.

[14] F. Weidner *et al.* incl. **C. Gerhardt**, "A Systematic Review on the Visualization of Avatars and Agents in AR & VR," *IEEE Trans. Vis. Comput. Graphics*, vol. 29, no. 5, pp. 2596–2606, 2023. doi:10.1109/TVCG.2023.324707

[15] **C. Gerhardt**, P. Aichroth, and S. Mann, "Selective Face Encryption in H.264 Encoded Videos," in *Proc. IEEE VCIP*, 2017, pp. 1–4. doi:10.1109/VCIP.2017.8305040.

Open-Source Projects & Datasets

AWARE	Weather-augmentation benchmark framework comprising three pipelines: **SWIFT** (augmentation), **PRISM** (generation), and **PROVE** (evaluation). `https://carhartt21.github.io/AWARE`
AWACS	Weather-balanced augmented dataset — 71 400 images across 25 augmentation strategies for robustness evaluation.
SkyCloud360	600 equirectangular panoramic images for sky and cloud segmentation. Code: `https://github.com/carhartt21/SkyCloud360` Data: `https://osf.io/a5ews`
SkyCloudNet	Sky and cloud segmentation network with pre-trained models. Code: `https://github.com/carhartt21/SkyCloudNet` Data: `https://osf.io/y69ah`
OUTSIDE15k	15 000 outdoor images with 24-class pixel-level annotations. Code & Data: `https://github.com/carhartt21/outsideNet`
360° Detection	2 500 panoramic images, 81 traffic sign classes. Code: `https://github.com/carhartt21/360-Object-Detection` Data: `https://osf.io/fsteu`

Technical Skills

Languages	Python, C/C++/C#, Java, MATLAB/Octave, SQL/MySQL, LaTeX
ML / DL	PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, ONNX
Tools	Git, Docker, Linux, Bash, Jupyter, Weights & Biases, Slurm
Domains	Semantic Segmentation, Object Detection, Image Augmentation, 360° Vision, VR/AR

Languages

German	Native
English	Fluent (C1+)
French	Basic knowledge
Serbian	Basic knowledge

Certifications

2023 **C-Lizenz (Coaching License)**
EU Drone Pilot Certificate (A1/A3)

Funding

2019 – 2024 Research funded by the Free State of Thuringia (Grant No. 2017 FE 9147), co-financed by the European Union under the European Regional Development Fund (ERDF).

Academic Service

Teaching	Courses in VR/AR and computer vision at TU Ilmenau
Supervision	Co-supervision of Bachelor's and Master's theses

Ilmenau, March 13, 2026